

Track-System – Glossary

Last updated: 23 Sep 2026 (after step 3) | The words that keep coming back in this project, each one in plain language | **[NEW]** = added in step 3

1. The big picture – who talks to whom

```
FRONTEND (in the browser)    BACKEND (on the computer)    DATABASE
React + Vite                 ASP.NET Core (C#)            SQL Server
|                             |                                     |
|--- axios sends HTTP req ---> endpoint /api/...           |
|                             |--- EF Core: C# to SQL -->    |
|<----- JSON answer -----|<----- rows -----|
```

The frontend NEVER talks to the database directly. Always through the backend.

2. The platform words

Word	What it means / where we meet it
.NET	Microsoft's platform for running C# programs, like Node.js is for JavaScript. It includes the compiler, the libraries and the runtime. (dotnet --version)
ASP.NET Core	The part of .NET for building web servers and APIs. Our whole backend.
SDK / Runtime	SDK = the tools to build code. Runtime = only what is needed to run it. A developer needs the SDK. We installed the SDK.
Kestrel	The actual web server inside ASP.NET Core. It listens on the port and receives the requests. You never write it, it is just there. (The line "Now listening on: ...")
NuGet	The library store for .NET. Every PackageReference in the .csproj is downloaded from there. (dotnet restore)
npm	The same idea for JavaScript. The frontend's package.json is its shopping list. (npm install)
Vite	The tool that runs the React project while developing and builds it for production. It gives the frontend its own address, usually port 5173. (npm run dev)

3. The web words

Word	What it means
HTTP request	A message the browser sends to a server. It carries a method (GET/POST/...), an address, optional headers and optional body data.
GET / POST / PUT / DELETE	GET = give me data. POST = here is new data (for example login details). PUT = update something existing. DELETE = remove it.
Endpoint	One address on the server plus the code that answers it. Example: /api/health.
API	The whole collection of endpoints. The menu of everything the backend can do for the frontend.
REST	A common style for designing an API: the addresses are nouns (/api/employees) and the methods are the verbs (GET, POST).
JSON	The text format both sides speak, for example {"status":"ok"}. C# objects become JSON on the way out, JSON becomes C# objects on the way in.

Word	What it means
Status code	A number describing the result: 200 OK, 201 created, 400 bad request, 401 not logged in, 403 not allowed, 404 address not found, 500 server crashed.
localhost	"This computer". Not reachable from outside, which is exactly what we want while developing.
Port	The door number on the machine. 5163 = our API, 5173 = React. Two programs cannot use the same port.
axios	A JavaScript library, already installed in the frontend, that sends the HTTP request from React and gives back the JSON answer.
CORS	A browser security rule: a page served from port 5173 may not call port 5163 unless the server says it is allowed. Adding that permission is a backend step.
[NEW] Connection refused	The browser's "can't reach this page": nothing is listening on that address. The server is not running or the port is wrong. It is not a code error.
[NEW] camelCase JSON	ASP.NET Core writes C# names (FirstName) into JSON as firstName automatically, the style JavaScript and React expect.

4. The database words

Word	What it means
SQL Server	Microsoft's database program. SQL Express is the free local edition.
SSMS	SQL Server Management Studio, the app where you look at the tables by hand.
Connection string	One line of text saying which server, which database and how to log in. Ours lives in appsettings.json.
Windows Authentication	Logging into the DB with your Windows user, so no password is written anywhere.
ORM	Object Relational Mapper: software that maps database rows to objects in code, so you stop writing SQL strings by hand.
EF Core	Entity Framework Core, Microsoft's ORM. It translates the C# we write into SQL and turns the returned rows back into C# objects.
DbContext	The EF Core class that represents the database. Our AppDbContext inherits from it, and every query goes through it.
DbSet	One line inside the DbContext representing one table, as a collection of objects you can query. Ours: DbSet<Employee> Employees (added in step 3).
Model / Entity	A plain C# class whose properties mirror the columns of one table. Ours: Models/Employee.cs. It is only a shape; EF Core fills it.
Migration	EF Core creating or changing tables out of your C# classes. We do NOT use it: our tables already exist, so we go the other way around.
Provider	The package that knows the dialect of one specific database. Ours is Microsoft.EntityFrameworkCore.SqlServer.
[NEW] Primary key	The column that identifies each row uniquely. In dbo.Employees it is eNum.
[NEW] IDENTITY	The DB fills the number automatically (1, 2, 3...). That is why we never send eNum when inserting.
[NEW] NULL / NOT NULL	NULL = the column may be empty. NOT NULL = a value is required.

Word	What it means
[NEW] DEFAULT	A value the DB fills in when none is given, for example role = 'Employee', status = 'pending'.
[NEW] Foreign key	A column pointing to another table's key. Employees.teamNum points to Teams.teamNum.
[NEW] Empty result []	An empty JSON list means the query worked but the table has no rows. An error looks different (red text or a 500).

5. The C# and code words

Word	What it means
Namespace	The address of a class so other files can find it with using Ours: TrackSystem.Api.Data, and now TrackSystem.Api.Models.
Class / Property	A class is a blueprint. Its properties are the fields inside it, which match the table columns.
Inheritance (:)	class AppDbContext : DbContext means it receives all of DbContext's abilities for free.
Lambda (=>)	A small function written inline and kept for later. The code inside MapGet is a lambda: it runs on each request, not at startup.
async / await	Talking to a DB takes time. await means "wait for the answer without freezing the server", so other requests keep being served.
Dependency Injection	You ask for something in the parentheses and the server hands you a ready one. That is why we never write new AppDbContext(...).
Service registration	Teaching the server how to build those objects. Anything registered with builder.Services must be written before Build().
Middleware	Code every request passes through on its way in (the CORS check, the login check, logging) before it reaches the endpoint.
Minimal API	The style we use: endpoints written directly in Program.cs. The older style is Controllers, which are separate class files.
[NEW] Attribute []	A label in square brackets that gives extra information about the ONE thing written right after it. Example: [Key].
[NEW] [Key]	Tells EF Core which property is the primary key. Needed on Enum because EF Core only guesses names like Id or EmployeeId.
[NEW] Enum vs enum	Enum is just our property name (employee number). enum is a C# keyword for a fixed list of values. C# is case-sensitive, so they never clash.
[NEW] Nullable ?	int? or string? means "may be empty". It is the C# side of a NULL column (teamNum, position, lockoutEnd).
[NEW] = string.Empty	A starting value of "" so a required text property is never null in C#.
[NEW] Types mapping	nvarchar → string, int → int, bit → bool, datetime → DateTime.
[NEW] LINQ	C#'s way of writing queries with methods like Select and Where. EF Core turns them into SQL.
[NEW] .Select(...)	Chooses which fields to take. EF Core reads only those columns, which is how the password never leaves the DB.
[NEW] Anonymous object	new { ... } with no class name: a quick object holding only the fields we list. Used to shape the JSON output.
[NEW] ToListAsync()	Actually runs the query and returns the rows as a list, without freezing the server.
[NEW] Results.Ok(...)	Returns HTTP 200 with the given object turned into JSON.

Word	What it means
[NEW] DTO	Data Transfer Object: a named class describing exactly what an endpoint sends. The planned replacement for the anonymous object.

6. The security words (coming in the next steps)

Word	What it means
Hashing	Turning a password into scrambled text that cannot be turned back. We compare hashes, so the real password is never stored.
BCrypt	The hashing library we will use. It is deliberately slow, which makes guessing attacks expensive.
Salt	Random characters added before hashing, so two people with the same password still get different hashes.
Authentication	Who are you? This is the login itself.
Authorization	What are you allowed to do? These are the permissions, for example the UserPermissions table.
JWT	JSON Web Token: a signed string the server gives you after a successful login. The frontend sends it back on every later request as proof of identity.
Token expiry	A JWT dies after a set time, so a stolen one is not useful forever.
[NEW] User Secrets	A .NET feature that stores secret settings (like the JWT key) on your own PC, outside the project folder, so they never reach the public GitHub repo.
[NEW] Never expose the hash	Even a hashed password is never sent to the browser. The model holds it; the endpoint leaves it out.

7. Tools and workflow

Word	What it means
git add / commit / push	add = choose the changes. commit = save a snapshot locally with a message. push = upload it to GitHub.
.gitignore	The list of files git must ignore: bin, obj, node_modules and anything secret.
bin / obj	Folders .NET generates while building. Never edit them, never commit them.
Build vs Run	Build = turn the C# into a program. Run = start it. dotnet run does both. dotnet build only checks it compiles.
Hot reload	React updates in the browser the moment you save. The backend does not: after every change you stop it with Ctrl+C and run it again.
[NEW] Working directory	The folder the terminal is "standing in", shown in the prompt. dotnet commands must run where the .csproj is: Backend\TrackSystem.Api.
[NEW] cd	Change directory: moves the terminal into another folder. Example: cd Backend\TrackSystem.Api.
[NEW] MSB1003	"Specify a project or solution file": dotnet found no .csproj in the current folder. Fix: cd into the project folder.
[NEW] Script Table as	SSMS menu (right-click a table → Script Table as → CREATE To) that writes the SQL describing an existing table. Read-only.